

Metamorphic rocks

Extreme pressure or heat can squeeze or cook rocks so much that they change their character, turning into metamorphic rocks. Rocks can be squeezed by tectonic forces, heated by intrusions of molten magma, or both. There are different grades of metamorphism, creating rocks that are harder and more crystalline at each stage.



MOUNTAIN SLATE

The forces that buckle sedimentary strata into mountain ranges also compress the rock itself. They can squeeze soft shale so much that the minerals break down and recrystallize in thin layers. The result is slate—a rock that is harder than shale, but which splits easily along the cleavage planes between layers. This makes it very useful for manufacturing thin, light, weatherproof roofing tiles.

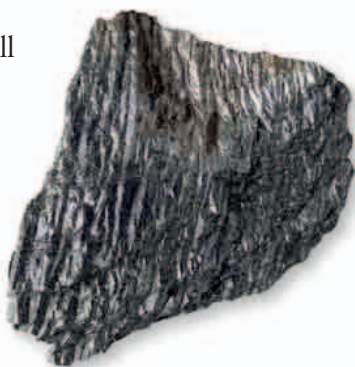
▲ SLATE QUARRY

The mountains of Snowdonia in Wales are an important source of high-quality roofing slate.

SPARKLING SCHIST

Increasing pressure and heat can turn slate and other rocks into a harder metamorphic rock called schist.

Like slate, schist contains flattened crystals that are all aligned the same way. This makes the crystal faces glitter with reflected light. Some types of schist also contain garnet crystals that grow from the original rock minerals.



BANDED GNEISS

In the subduction zones that destroy the edges of tectonic plates, intense pressure and extreme heat create a highly compressed metamorphic rock called gneiss. It usually has dark and pale bands of crystals that are crumpled and distorted. The oldest-known rocks on the planet are gneisses that were created almost 4 billion years ago.

